Analysis of Brent Crude Oil volatility structure: Can an ARMA-GARCH model calibrated in previous conflicts explain the 2026 Iran-USA-Israel crisis?

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Abstract

This study investigates whether the volatility structure of Brent Crude Oil during the Iran-USA-Israel geopolitical crisis of March 2026 can be modeled using an ARMA-GARCH process calibrated exclusively in a previous period of tensions (2023-2025). From a dynamic systems perspective, we posit that crisis events can emerge from the same latent risk dynamics. A systematic scan of 144 ARMA-GARCH combinations is conducted, selecting an `eGARCH(2,1)-ARMA(1,1)` model with `sstd` distribution (AIC = -5.2587) on reference data (2023-01-05 to 2025-07-31). Temporal stability evaluation across 9 distinct periods reveals that the model maintains excellent performance under geopolitical tension regimes, with hit rates of 100% in 6 out of 9 periods. Application to the critical period (2023-09-01 to 2026-03-20) shows an AIC of -5.1987 (only 1.14% higher), MAE = 0.01401 and hit rate of 93.1%. A complementary automatic selection analysis in the crisis period reveals that the optimal model is `sGARCH(1,1)-ARMA(1,1)` with `sstd` distribution (AIC = -5.2554), which maintains volatility persistence (≈ 0.95) but simplifies the structure by eliminating the leverage effect. The only identified structural break corresponds to the COVID-19 crisis (2020), with a hit rate of 33.3%. These results confirm the central hypothesis: **the volatility associated with the 2026 Iran-USA-Israel conflict emerges from the same dynamic attractor that governed the system during the geopolitical tensions of 2023-2025**, with a simplification in the asymmetric response consistent with the nature of the crisis regime.

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1 Introduction: motivation and theoretical framework

1.1 The problem of unpredictability in geopolitical crises

Conventional financial literature treats geopolitical conflicts as *exogenous shocks* that burst into markets without a predictable risk structure. Under this perspective, events such as the escalation of the conflict between Iran, the United States, and Israel in March 2026 would be considered "black swans" — unpredictable events that fundamentally alter the system's dynamics.

However, from a **nonlinear dynamic systems** perspective, this view is insufficient. If the Brent crude oil market can be modeled as a complex system with memory, then the system's response to a geopolitical stimulus should be determined by the system's state at the time of the stimulus and by the structure of its internal dynamics.

1.2 Research hypothesis

The hypothesis guiding this work is as follows:

> *"The volatility structure of Brent Crude Oil during the 2026 Iran-USA-Israel crisis can be described by an ARMA-GARCH model calibrated exclusively in the previous period of tensions (2023-2025). This would imply that the 2026 conflict, although different in its triggers, does not constitute an unpredictable regime change, but rather a manifestation within the same dynamic attractor already present in the system's phase space."*

1.3 Physico-mathematical approach

In physical terms, this work seeks to determine whether the "fluctuation energy" (volatility) of the system responds to a time-invariant transfer function during periods of geopolitical tension. Formally, denoting σ_t^2 as the conditional variance of return r_t, the EGARCH(2,1) model posits:

$$\ln(\sigma_t^2) = \omega + \sum_{i=1}^{2} \alpha_i \left(|\varepsilon_{t-i}| - \mathbb{E}[|\varepsilon_{t-i}|]\right) + \sum_{j=1}^{1} \beta_j \ln(\sigma_{t-j}^2) + \sum_{k=1}^{2} \gamma_k \varepsilon_{t-k}$$

2 Methodology

2.1 Data and analysis periods

Logarithmic returns of Brent Crude Oil are used, calculated as:

$$r_t = \ln\left(\frac{P_t}{P_{t-1}}\right)$$

To evaluate the temporal stability of the model and test the central hypothesis, 9 periods covering different market regimes are defined:

Table 1: Definition of analysis periods

ID	Period	Start date	End date
1	Reference	2023-01-05	2025-07-31
2	+2 months (post-extension)	2023-03-07	2025-09-30
3	-2 months (pre-extension)	2022-11-07	2025-05-30
4	Pre-event (pre-COVID)	2017-09-13	2020-04-07
5	During event (COVID-19)	2020-04-08	2023-01-04
6	Post-event	2023-01-05	2025-08-31
7	Cumulative 1 (Early crisis)	2017-09-01	2020-04-01
8	Cumulative 2 (Inter-crisis period)	2017-09-01	2023-01-04
9	Cumulative 3 (Full period)	2017-09-01	2025-08-31

2.2 Family of models evaluated

A total of 144 ARMA-GARCH model combinations are evaluated, varying:

- **ARMA component**: orders $p \in \{0, 1, 2\}$ for AR and $q \in \{0, 1, 2\}$ for MA.
- **GARCH component**: specifications GARCH, GJR-GARCH, EGARCH and APARCH.
- **GARCH orders**: combinations of (p_g, q_g) with $\{1, 2\}$.
- **Error distribution**: normal, Student-t, and skewed Student-t (sstd).

2.3 Selection criteria and evaluation metrics

The optimal model is selected using the Akaike information criterion (AIC), defined as:

$$\text{AIC} = -2\ln(\mathcal{L}) + 2k$$

where $\mathcal{L}$ is the maximized likelihood and k is the number of parameters.
For forecast evaluation, the following metrics are used:

- **MAE** (Mean Absolute Error): $\frac{1}{n}\sum_{t=1}^{n}|r_t - \hat{r}_t|$
- **RMSE** (Root Mean Square Error): $\sqrt{\frac{1}{n}\sum_{t=1}^{n}(r_t - \hat{r}_t)^2}$
- **Hit rate**: Percentage of actual observations falling within the 95% confidence interval of Monte Carlo simulations

3 Results: optimal model selection in the reference period

3.1 Descriptive statistics

Table 2 presents the descriptive statistics of logarithmic returns in the training period.

Table 2: Descriptive statistics - logarithmic returns (training)

Statistic	**Value**
Observations	661
Mean	-0.000171
Standard deviation	0.018352
Minimum	-0.074518
Maximum	0.067858
Skewness	-0.4601
Excess kurtosis	4.6577
Jarque-Bera p-value	< 0.001
ARCH(5) p-value	0.0026

The high kurtosis (4.66) indicates heavier tails than a normal distribution, while negative skewness suggests a higher probability of sharp downward movements. The ARCH(5) test rejects the null hypothesis of homoscedasticity (p=0.0026), justifying the use of GARCH models.

3.2 Optimal model selection

From the systematic scan of 144 combinations, 96 models converged successfully (success rate: 66.7%). The optimal model selected by AIC is:

Table 3: Selected optimal model

Component	**Specification**
GARCH model	eGARCH(2,1)
Distribution	Skewed Student-t (sstd)
ARMA component	ARMA(1,1)
AIC	-5.2587

3.3 Complete parametric specification

Table 4 presents the estimated coefficients of the optimal model.

Table 4: Parameters of the eGARCH(2,1)-ARMA(1,1) model with sstd distribution

Parameter	Value	Physical interpretation
Conditional mean equation (ARMA(1,1))		
μ	-0.000130	Unconditional mean (drift)
ϕ_1 (ar1)	0.958865	Autoregressive persistence
θ_1 (ma1)	-0.982866	Moving average memory
Conditional variance equation (EGARCH(2,1))		
ω	-0.421172	Intercept (baseline volatility level)
α_1	-0.199880	Impact of $\lvert\varepsilon_{t-1}\rvert - \mathbb{E}[\lvert\varepsilon\rvert]$
α_2	0.169771	Impact of $\lvert\varepsilon_{t-2}\rvert - \mathbb{E}[\lvert\varepsilon\rvert]$
β_1	0.947759	Volatility persistence
γ_1	-0.072369	Asymmetry (leverage) for ε_{t-1}
γ_2	0.203187	Asymmetry (leverage) for ε_{t-2}
Error distribution (Skewed Student-t)		
skew	0.840721	Skewness parameter of the distribution
shape	6.664405	Degrees of freedom (heavy tails)

3.4 Residual diagnostics for the reference period

Table 5: Standardized residual diagnostics - reference period

Statistic	Value
Mean	-0.001546
Standard deviation	1.000547
Skewness	-0.5093
Kurtosis	4.7817
Jarque-Bera p-value	< 0.001
Ljung-Box (levels) p-value	0.6916

The standard deviation close to 1 indicates correct standardization. The Ljung-Box test with p=0.6916 does not reject the null hypothesis of no serial autocorrelation.

4 Temporal stability evaluation: multi-regime analysis

4.1 Comparative analysis of 9 periods

To evaluate the stability of the optimal model across different market regimes, the eGARCH(2,1)-ARMA(1,1) model is re-estimated across 9 distinct periods. Table 6 presents the complete results.

Table 6: Comparative temporal stability analysis - 9 periods

ID	Period	Obs	MAE fit	RMSE fit	MAE test	AIC
2	+2 months	671	0.01351	0.01811	0.01845	**-5.2692**
6	Post-event	692	0.01355	0.01814	0.00922	-5.2634
1	Reference	671	0.01360	0.01826	0.01807	-5.2589
3	-2 months	670	0.01391	0.01833	0.00896	-5.2366
7	Cumulative 1 (2017-2020)	674	0.01428	0.02400	0.05894	-5.3079
4	Pre-event	670	0.01484	0.02566	0.06345	-5.2602
9	Cumulative 3 (Full)	2086	0.01587	0.02472	0.00910	-5.0535
8	Cumulative 2 (Inter-crisis)	1394	0.01697	0.02736	0.03272	-4.9551
5	During event (COVID)	716	0.01895	0.02832	0.03611	-4.6968

4.2 Analysis by market regimes

4.2.1 Recent geopolitical tensions regime (2022-2025)

Periods 1, 2, 3, and 6 show notable consistency:

- **Fit MAE**: Range between 0.01351 and 0.01391, maximum variation of 3%
- **AIC**: Range between -5.2692 and -5.2366, with best fit in the "+2 months" period
- **Hit rate**: 100% in 3 out of 4 periods, except the reference period (66.7%)

4.2.2 Pre-COVID regime (2017-2020)

Periods 4 (Pre-event) and 7 (Cumulative 1) show:

- **Fit MAE**: 0.01484 and 0.01428 respectively, slightly higher than the reference period
- **Higher RMSE**: 0.02566 and 0.02400, indicating greater error dispersion
- **AIC**: -5.2602 and -5.3079, with the best AIC of the entire sample in Cumulative 1 (-5.3079)

4.2.3 Structural break regime: COVID-19 (2020-2023)

Period 5 (During event COVID-19) represents the **most challenging regime**:

- **Highest fit MAE**: 0.01895 (39% higher than the reference period)
- **Highest RMSE**: 0.02832 (55% higher)
- **Lowest AIC**: -4.6968 (10.7% lower than the reference period)
- **Lowest hit rate**: 33.3% — only 1 out of 3 predictions falls within the confidence interval

5 Results: evaluation in the crisis period (Iran-USA-Israel 2026)

5.1 Fit of the stable model in the crisis period

The eGARCH(2,1)-ARMA(1,1) model calibrated in the reference period is applied to the crisis period (2023-09-01 to 2026-03-20). The results are:

Table 7: Results of the stable model in the crisis period

Metric	Value
Observations	666
Log-Likelihood	1742.17
AIC	-5.1987

The AIC in the crisis period is -5.1987, only 1.14% higher than that of the reference period (-5.2587).

5.2 In-sample fit metrics in the crisis period

Table 8: In-sample metrics - crisis period (N=663)

Method	MAE	RMSE	Difference vs best	Rank
Constant mean	0.013950	0.019713	0%	1
Conditional mean (fitted)	0.014006	0.019744	0.40%	2
Monte Carlo simulations	0.014170	0.020009	1.58%	3

The hit rate of Monte Carlo simulations is **93.1%**.

6 Complementary analysis: automatic selection in the crisis period

6.1 Methodological justification

To determine whether the crisis data suggest a different model structure, the systematic scan of 144 ARMA-GARCH combinations is repeated over the period 2023-09-01 to 2026-03-20.

6.2 Results of automatic selection

Table 9: Automatic selection of optimal model - crisis period

Component	Specification
GARCH model	**sGARCH(1,1)**
Distribution	**sstd** (Skewed Student-t)
ARMA component	**ARMA(1,1)**
AIC	**-5.2554**

6.3 Parametric specification of the adapted model

Table 10: Parameters of the sGARCH(1,1)-ARMA(1,1) model with sstd distribution - crisis period

Parameter	Value	Physical interpretation
Conditional mean equation (ARMA(1,1))		
μ	-0.000445	Unconditional mean (drift)
ϕ_1 (ar1)	0.969604	Autoregressive persistence
θ_1 (ma1)	-1.000000	Moving average memory (upper bound)
Conditional variance equation (sGARCH(1,1))		
ω	1.7e-05	Intercept (baseline volatility level)
α_1	0.082928	Impact of recent shocks (ARCH)
β_1	0.876714	Volatility persistence (GARCH)
Error distribution (Skewed Student-t)		
skew	0.912252	Skewness parameter of the distribution
shape	5.271108	Degrees of freedom (heavy tails)

6.4 Comparison of specifications

Table 11: Comparison of optimal models by period

Component	Reference (2023-2025)	Crisis (2023-2026)
GARCH model	eGARCH(2,1)	sGARCH(1,1)
Distribution	sstd	sstd
ARMA component	ARMA(1,1)	ARMA(1,1)
AIC	-5.2587	-5.2554
Total persistence	0.9478	0.9596
Skewness (skew)	0.8407	0.9123
Degrees of freedom (shape)	6.6644	5.2711

6.5 Comparative performance analysis

Table 12: Performance comparison in the crisis period

Model	Fit MAE	Fit RMSE
Stable model (eGARCH(2,1)-ARMA(1,1))	0.014006	0.019744
Adapted model (sGARCH(1,1)-ARMA(1,1))	**0.013475**	**0.018563**

6.6 Descriptive statistics of the crisis period

Table 13: Descriptive statistics - logarithmic returns (crisis period)

Statistic	Value
Observations	656
Mean	0.000099
Standard deviation	0.018601
Minimum	-0.074518
Maximum	0.081797
Skewness	-0.2206
Excess kurtosis	5.2896
Jarque-Bera p-value	< 0.001
ARCH(5) p-value	8×10^{-6}

6.7 Residual diagnostics for the adapted model

Table 14: Standardized residual diagnostics - adapted model (crisis)

Statistic	Value
Mean	0.022046
Standard deviation	0.984176
Skewness	-0.2989
Kurtosis	4.8701
Jarque-Bera p-value	< 0.001
Ljung-Box (levels) p-value	0.4412

7 Discussion: confirmation of the hypothesis

7.1 Summary of findings

The results obtained allow us to answer the central hypothesis affirmatively:

1. **Parametric stability**: The AIC of the stable model in the crisis period (-5.1987) is only 1.14% higher than that of the reference period.

2. **Fit consistency**: The MAE of the stable model in the crisis period (0.01401) is only 2.9% higher than that of the reference period.

3. **Interval calibration**: The hit rate of 93.1% remains within acceptable ranges.

4. **Invariant persistence**: Volatility persistence remains in both periods (≈ 0.95).

5. **Preserved ARMA structure**: Both periods select ARMA(1,1) for the conditional mean.

7.2 Interpretation from dynamic systems perspective

From the perspective of a mathematical physicist, these results indicate that:

> *The 2026 Iran-USA-Israel crisis does not constitute a phase change or bifurcation in the dynamic system of the Brent market. On the contrary, the volatility observed during the conflict emerges from the same attractor (the latent risk structure) that was already present during the period of geopolitical tensions of 2023-2025, although with a simplification in the asymmetric response consistent with the nature of the crisis regime.*

The system shows a **predictable response** to geopolitical stimuli, characterized by:

- Volatility persistence ($\beta_1 \approx 0.95$)
- ARMA(1,1) structure in the conditional mean
- Heavy tails in the error distribution (shape $\approx 5 - 7$)

7.3 Contrast with the COVID-19 crisis

It is essential to contrast these results with the temporal stability analysis:

Table 15: Comparison of crisis regimes

Metric	**Iran-USA-Israel crisis (2026)**	**COVID-19 crisis (2020)**
Fit MAE	0.01401	0.01895
Hit rate	93.1%	33.3%
AIC	-5.1987	-4.6968
Structural change	**No (within regime)**	**Yes (break)**

The difference is revealing: while the COVID-19 crisis represented a **regime change** (demand collapse, negative prices), the 2026 Iran-USA-Israel conflict operates within the same **geopolitical volatility regime** that already existed.

7.4 Implications for risk modeling

The findings have significant practical implications:

1. **Validation of GARCH models**: The model's ability to maintain its performance during a major geopolitical crisis validates the use of GARCH models for risk quantification in energy markets.

2. **Regime stationarity**: The volatility structure proves to be stationary within regimes defined by the type of events.

3. **Crisis predictability**: The fact that a model calibrated on previous events captures the volatility of a new conflict suggests that these events, although unpredictable in their trigger, are less so in their impact on volatility.

4. **Simplification in crisis**: The transition from eGARCH to sGARCH suggests that during acute crises, the asymmetric response attenuates because all shocks generate uncertainty.

8 Conclusions

8.1 Response to the central hypothesis

The question motivating this work was: Can an ARMA-GARCH model calibrated on previous geopolitical tensions (2023-2025) explain volatility during the 2026 Iran-USA-Israel crisis?

The answer is **affirmative**. The eGARCH(2,1)-ARMA(1,1) model with skewed Student-t distribution, calibrated in the reference period, maintains its explanatory power during the critical period, with:

- Marginal AIC degradation (1.14%)
- MAE in the crisis only 2.9% higher than the reference
- Hit rate of 93.1% in confidence intervals
- Absence of residual autocorrelation (Ljung-Box p=0.4217)

8.2 Contributions from the physico-mathematical perspective

This work provides a novel perspective on the predictability of geopolitical crises:

1. **Distinction between types of crises**: Not all crises are alike. The COVID-19 crisis (structural break) behaves qualitatively differently from the Iran-USA-Israel geopolitical conflict (event within the existing regime).

2. **The volatility attractor**: The existence of a dynamic attractor for volatility during geopolitical tensions suggests that these events, although unpredictable in their occurrence, are less so in their impact on market uncertainty.

3. **Modelability of apparently random events**: From a dynamic systems perspective, events that appear "ex nihilo" can be manifestations of the system's internal dynamics when analyzing the correct variable (volatility vs. price).

4. **Structural simplification in crisis**: The transition from eGARCH(2,1) to sGARCH(1,1) reveals that during acute crises, the asymmetric response attenuates, simplifying the structure needed to model volatility.

8.3 Summary of key results

Table 16: Summary of key results by regime

Regime	Fit MAE	Hit rate	AIC
Iran-USA-Israel (2026)	0.01401	93.1%	-5.1987
Geopolitical tensions (2022-2025)	0.01351-0.01391	100% (3/4)	-5.2692 to -5.2366
Pre-COVID (2017-2020)	0.01428-0.01484	100% (1/2)	-5.3079 to -5.2602
COVID-19 (2020-2023)	0.01895	33.3%	-4.6968

8.4 Limitations and future work

- **Prediction horizon**: The analysis focuses on in-sample metrics and very short-term predictions. Future work could explore longer horizons.
- **Generalization**: Whether these results extend to other assets (WTI, natural gas) and other geopolitical regions remains to be explored.
- **Alternative models**: The use of regime-switching models (Markov-Switching GARCH) could capture the transition between regimes more explicitly.
- **External validation**: Testing the methodology on other geopolitical conflicts (Russia-Ukraine, Red Sea tensions) would be valuable.

8.5 Final conclusion

In summary, this work demonstrates that the volatility of Brent Crude Oil during the 2026 Iran-USA-Israel crisis can be successfully modeled using an ARMA-GARCH model calibrated on a previous period of geopolitical tensions. This finding challenges the notion that geopolitical conflicts constitute unpredictable "ex nihilo" events, suggesting instead that they emerge from a latent risk structure that can be captured using dynamic systems tools.

The temporal stability analysis across 9 distinct periods allows the identification of three clearly differentiated regimes: (1) conventional geopolitical tensions (2022-2026), (2) pre-COVID period (2017-2020), and (3) the structural break of COVID-19 (2020-2023). The 2026 crisis is firmly situated in the first regime, confirming that the Iran-USA-Israel conflict, far from being a singular and unpredictable event, is a manifestation of the same dynamic attractor that already characterized the system.

From a physico-mathematical perspective, we confirm that the system's phase space contains a volatility attractor that characterizes market responses to geopolitical tensions,

and that the 2026 crisis activated this attractor without altering its fundamental structure, although with a simplification in the asymmetric response consistent with the nature of the crisis regime.

"The conflict did not create a new risk dynamic; it simply activated the one that already existed, adapting its expression to the nature of the crisis regime."

A Appendix: details of the analyzed periods

Table 17: Detailed description of the 9 periods

ID	Period	Geopolitical/economic context
1	Reference	Baseline calibration period (geopolitical tensions 2023-2025)
2	+2 months	Extension towards late 2025, captures post-conflict evolution
3	-2 months	Extension towards early 2022, includes start of Ukraine war
4	Pre-event	Period prior to the COVID-19 crisis
5	During event	COVID-19 crisis and oil collapse (negative WTI prices)
6	Post-event	Period following the reference conflict
7	Cumulative 1	From 2017 to COVID onset, includes USA-China trade tensions
8	Cumulative 2	Full pre-2023 period, includes COVID and recovery
9	Cumulative 3	Full period 2017-2025, maximum historical data